Bi-directional Programmable DC Power Supply

IT6600C Programming Guide (Battery Emulator)



Model: IT6600C Series

Version: V1.0

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⎓	Direct current	▮	ON (power on)
∼	Alternating current	○	OFF (power off)
⎓ ∼	Both direct and alternating current	⎍	Power-on state
⏚	Protective conductor terminal	⎎	Power-off state
⏚	Earth (ground) terminal	±	Reference terminal
⚠	Caution, risk of electric shock	+	Positive terminal

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	Warning, risk of danger (refer to this manual for specific Warning or Caution information)	▬	Negative terminal
	Frame or chassis terminal	-	-

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- Do not use the instrument if it is damaged. Before operation, check the casing to see whether it cracks. Do not operate the instrument in the presence of inflammable gasses, vapors or dusts.
- The instrument is provided with power cord during delivery and should be connected to the distribution box. Before operation, be sure that the instrument is well grounded.
- Make sure to use the power cord supplied by ITECH.
- Check all marks on the instrument before connecting the instrument to power supply.
- Use electric wires of appropriate load. All loading wires should be capable of bearing maximum short-circuit of electronic load without overheating. If there are multiple loads, each pair of the load power cord must be carry out the full rated short-circuit output current of the power securely.
- Ensure the voltage fluctuation of mains supply is less than 10% of the working voltage range in order to reduce risks of fire and electric shock.
- Do not install alternative parts on the instrument or perform any unauthorized modification.
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CAUTION

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- Always clean the casing with a dry cloth. Do not clean the internals.
- Make sure the vent hole is always unblocked.

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The instrument is designed for indoor use and an area with low condensation. The table below shows the general environmental requirements for the instrument.

Environmental Conditions	Requirements
Operating temperature	0°C to 40°C
Operating humidity	20%-80% (non-condensation)
Storage temperature	-20°C to 70 °C
Altitude	Operating up to 2,000 meters
Pollution degree	Pollution degree 2
Installation category	II

NOTE

To make accurate measurements, allow the instrument to warm up for 30 min before operation.

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CE	The CE tag shows that the product complies with the provisions of all relevant European laws (if the year is shown, it indicates that the year when the design is approved).
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2002/96/EC Waste Electrical and Electronic Equipment (WEEE) Directive

This product complies with the WEEE Directive (2002/96/EC) marking requirement. This affix product label indicates that you must not discard the electrical/electronic product in domestic household waste.

Product Category

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Complies with the essential requirements of the following applicable European Directives, and carries the CE marking accordingly:

- Electromagnetic Compatibility (EMC) Directive 2014/30/EU
- Low-Voltage Directive (Safety) 2014/35/EU

Conforms with the following product standards:

EMC Standard

IEC 61326-1 / EN 61326-1 [123]

Reference Standards

CISPR 11, Class A
IEC 61000-3-2
IEC 61000-3-3
IEC 61000-4-2
IEC 61000-4-3
IEC 61000-4-4
IEC 61000-4-5
IEC 61000-4-6
IEC 61000-4-11

Chapter 1 The product is intended for use in non-residential/non-domestic environments. Use of the product in residential/domestic environments may cause electromagnetic interference.

Chapter 2 Connection of the instrument to a test object may produce radiations beyond the specified limit.

Chapter 3 Use high-performance shielded interface cable to ensure conformity with the EMC standards listed above.

Safety Standard

IEC 61010-1:2010+A1:2019

Note

This section provides general and universal terms. For more detailed information on the Declaration of Conformity, please contact ITECH personnel.

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Chapter 1 SCPI Introduction

1.1 Overview

SCPI is short for Standard Commands for Programmable Instruments which defines a communication method of bus controller and instrument. It is based on ASCII and supply for testing and measuring instruments. SCPI command is based on hierarchical architecture which also known as tree system. In this system, Relevant Command is returned to a common node or root, so that a subsystem is formed. A part of OUTPut subsystem is listed below:

OUTPut:

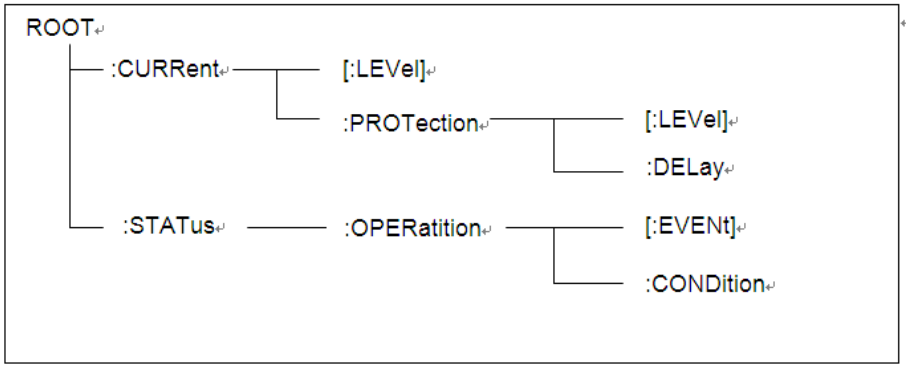
- **SYNC {OFF|0|ON|1}**
- **SYNC:**
  - **MODE {NORMal|CARRier}**
  - **POLarity {NORMal|INVerted}**

OUTPut is the root class keyword, SYNC is the second keyword, MODE and POLarity are the third keyword. Colon(:) is used for separating the command keyword and the next level keyword.

1.2 Command Type of SCPI

SCPI has two types of commands, common and subsystem.

- Common commands generally are not related to specific operation but to controlling overall instrument functions, such as reset, status, and synchronization. All commoncommands consist of a three-letter mnemonic preceded by an asterisk: ***RST *IDN? *SRE 8**.
- Subsystem commands perform specific instrument functions. They are organized into an inverted tree structure with the "root" at the top. The following figure shows a portion of a subsystem command tree, from which you access the commands located along the various paths.



Multiple Commands in a Message

Multiple SCPI commands can be combined and sent as a single message with one message terminator. There are two important considerations when sending several commands within a single message:

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- Use a semicolon to separate commands within a message.
- Head paths influence how the instrument interprets commands.

We consider the head path as a string which will be inserted in front of every command of a message. As for the first command of a message, the head path is a null string; for each subsequent command, the head path is a string which is defined to form the current command until and including the head of the last colon separator. A message with two combined commands:

CURR:LEV 3;PROT:STAT OFF

The example indicates the effect of semicolon and explains the concept of head path. Since the head path is defined to be "CURR" after "curr: lev 3", the head of the second command, "curr", is deleted and the instrument explains the second command as:

CURR:PROT:STAT OFF

If "curr" is explicitly included in the second command, it is semantically wrong. Since combining it with the head path will become "CURR:CURR:PROT:STAT OFF", resulting in wrong command.

Movement in the Subsystem

In order to combine commands from different subsystems, you need to be able to reset the header path to a null string within a message. You do this by beginning the command with a colon (:), which discards any previous header path. For example, you could clear the output protection and check the status of the Operation Condition register in one message by using a root specifier as follows:

PROTection:CLEAr;:STATus:OPERation:CONDition?

The following message shows how to combine commands from different subsystems as well as within the same subsystem:

POWer:LEVel 200;PROTection 28; :CURRent:LEVel 3;PROTection:STATe ON

Note the use of the optional header LEVel to maintain the correct path within the voltage and current subsystems, and the use of the root specifier to move between subsystems.

Including Common Commands

You can combine common commands with subsystem commands in the same message. Treat the common command as a message unit by separating it with a semicolon (the message unit separator). Common commands do not affect the header path; you may insert them anywhere in the message.

VOLTage:TRIGgered 17.5;:INITialize;*TRG

OUTPut OFF;*RCL 2;OUTPut ON

Case Sensitivity

Common commands and SCPI commands are not case sensitive. You can use upper or lower, for example:

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***RST = *rst**

:DATA? = :data?

:SYSTem:PRESet = :system:preset

Long-Form and Short-Form Versions

A SCPI command word can be sent in its long-form or short-form version. However, the short-form version is indicated by upper case characters. Examples:

:SYSTem:PRESet long-form

:SYST:PRES short form

:SYSTem:PRES long-form and short-form combination

Note that each command word must be in long-form or short-form, and not something in between.

For example, **:SYSTe:PRESe** is illegal and will generate an error. The command will not be executed.

Query

Observe the following precautions with queries:

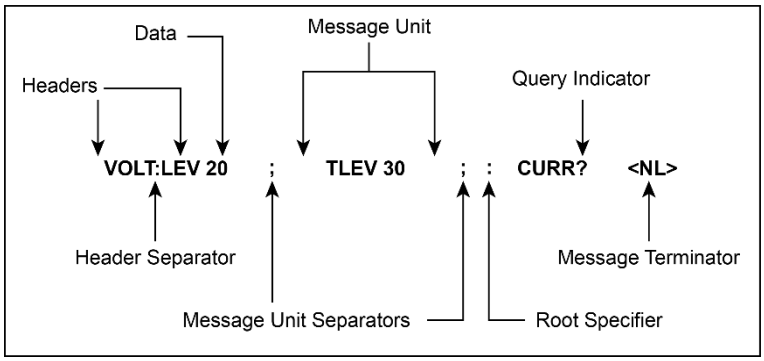
- Set up the proper number of variables for the returned data. For example, if you are reading back a measurement array, you must dimension the array according to the number of measurements that you have placed in the measurement buffer.
- Read back all the results of a query before sending another command to the instrument. Otherwise a Query Interrupted error will occur and the unreturned data will be lost.

1.3 Message Type of SCPI

There are two types of SCPI messages, program and response.

- Program message: A program message consists of one or more properly formatted SCPI commands sent from the controller to the instrument. The message, which may be sent at any time, requests the instrument to perform some action.
- Response message: A response message consists of data in a specific SCPI format sent from the instrument to the controller. The instrument sends the message only when commanded by a program message called a "query."

The next figure illustrates SCPI message structure:



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The message unit

The simplest SCPI command is a single message unit consisting of a command header (or keyword) followed by a message terminator. The message unit may include a parameter after the header. The parameter can be numeric or a string.

ABORt<NL>

VOLTage 20<NL>

Headers

Headers, also referred to as keywords, are instructions recognized by the instrument. Headers may be either in the long form or the short form. In the long form, the header is completely spelled out, such as VOLTAGE, STATUS and DELAY. In the short form, the header has only the first three or four letters, such as VOLT, STAT and DEL.

Query indicator

Following a header with a question mark turns it into a query (**VOLTage?**, **VOLTage:PROTection?**). If a query contains a parameter, place the query indicator at the end of the last header (**VOLTage:PROTection?MAX**).

Message unit separator

When two or more message units are combined into a compound message, separate the units with a semicolon (**STATus:OPERation?;QUEStionable?**).

Root specifier

When it precedes the first header of a message unit, the colon becomes the root specifier. It tells the command parser that this is the root or the top node of the command tree.

Message terminator

A terminator informs SCPI that it has reached the end of a message. Three permitted message terminators are:

- newline (<NL>), decimal 10 or hexadecimal 0X0A in ASCII.
- end or identify (<END>)
- both of the above (<NL><END>).

In the examples of this guide, there is an assumed message terminator at the end of each message.

Command execution rules

- Commands execute in the order that they are presented in the program message.
- An invalid command generates an error and, of course, is not executed.
- Valid commands that precede an invalid command in a multiple command program message are executed.
- Valid commands that follow an invalid command in a multiple command program message are ignored.

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1.4 Response Data Type

Character strings returned by query statements may take either of the following forms, depending on the length of the returned string:

- **<CRD>**: character response data. Permits the return of character strings.
- **<AARD>**: arbitrary ASCII response data. Permits the return of undelimited 7-bit ASCII. This data type has an implied message terminator.
- **<SRD>**: string response data. Returns string parameters enclosed in double quotes.
- **<Block>**: arbitrary block data.

Response messages

A response message is the message sent by the instrument to the computer in response to a query command.

Sending a response message

After sending a query command, the response message is placed in the Output Queue. When the instrument is then addressed to talk, the response message is sent from the Output Queue to the computer

Multiple response messages

If you send more than one query command in the same program message, the multiple response messages for all the queries is sent to the computer when the instrument is addressed to talk. The responses are sent in the order that the query commands were sent and are separated by semicolons (;). Items within the same query are separated by commas (,). The following example shows the response message for a program message that contains four single item query commands:

0; 1; 1; 0

Response message terminator (RMT)

Each response is terminated with an LF (line feed) and EOI (end or identify). The following example shows how a multiple response message is terminated:

0; 1; 1; 0; <RMT>

Message exchange protocol

Two rules summarize the message exchange protocol:

- **Rule 1**: You must always tell the instrument what to send to the computer.

  The following two steps must always be performed to send information from the instrument other computer:
  1. Send the appropriate query command(s) in a program message.
  2. Address the instrument to talk.
- **Rule 2**: The complete response message must be received by the computer before another program message can be sent to the instrument.

1.5 Command Format

Formats for command display are as follows:

[SOURce[1|2]:]VOLTage:UNIT {VPP|VRMS|DBM}

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[SOURce[1|2]:]FREQuency:CENTer
{<frequency>|MINimum|MAXimum|DEFault}

Based on the command syntax, most commands (and certain Parameter) are expressed in both upper and lower cases. Upper case refers to abbreviation of commands. Shorter program line may send commands in abbreviated format. Long-format commands may be sent to ensure better program readability.

For example, both formats of VOLT and VOLTAGE are acceptable in the above syntax statements. Upper or lower case may be used. Therefore, formats of VOLTAGE, volt and Volt are all acceptable. Other formats (such as VOL and VOLTAG) are invalid and will cause errors.

- Parameter options with given command strings are included in the brace ({ }). The brace is not sent along with command strings.
- Vertical stripes (|) separate several parameter options with given command strings. For example, {VPP|VRMS|DBM} indicates that you may assign "APP", "VRMS" or "DBM" in the above commands. Vertical stripes are not sent along with command strings.
- Angle brackets (< >) in the second example indicates that a value must be assigned to the parameter in the brace. For example, the parameter in the angle bracket is <frequency> in the above syntax statements. Angle brackets are not sent along with command strings. You must assign a value (such as "FREQ:CENT 1000") to the parameter, unless you select other options displayed in the syntax (such as "FREQ:CENT MIN").
- Some syntax elements (such as nodes and Parameter) are included in square brackets ([ ]). It indicates that these elements can be selected and omitted. Angle brackets are not sent along with command strings. If no value is assigned to the optional Parameter, the instrument will select a default value. In the above examples, "SOURce[1|2]" indicates that you may refer to source channel 1 by "SOURce" or "SOURce1" or "SOUR1" or "SOUR". In addition, since the whole SOURce node is optional (in the square bracket), you can refer to the channel 1 by omitting the whole SOURce node. It is because the channel 1 is the default channel for SOURce language node. On the other hand, if you want to refer to channel 2, "SOURce2" or "SOUR2" must be used in the program line.

Colon (:)

It is used to separate key words of a command with the key words in next level. As shown below:

APPL:SIN 455E3,1.15,0.0

In this example, APPLy command assigns a sine wave with frequency of 455 KHz, amplitude of 1.15 V and DC offset of 0.0 V.

Semicolon (;)

It is used to separate several commands in the same subsystem and can also minimize typing. For example, to send the following command string:

TRIG:SOUR EXT; COUNT 10

has the same effect as sending the following two commands:

TRIG:SOUR EXT
TRIG:COUNT 10

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Question mark (?)

You can insert question marks into a command to query current values of most Parameter. For example, the following commands will trigger to set the count as 10:

TRIG:COUN 10

Then, you may query count value by sending the following command:

TRIG:COUN?

You may also query the allowable minimum or maximum count as follows:

TRIG:COUN?MIN
TRIG:COUN?MAX

Comma (,)

If a command requires several Parameter, then a comma must be used to separate adjacent Parameter.

Space

You must use blank characters, [TAB] or [Space] to separate Parameter with key words of commands.

Common commands (*)

The IEEE-488.2 standard defines a set of common commands that perform functions such as reset, self-test, and status operations. Common commands always start with a asterisk (*) and occupy 3 character sizes, including one or more Parameter. Key words of a command and the first parameter are separated by a space. Semicolon (;) can separate several commands as follows:

***RST; *CLS; *ESE 32; *OPC?**

Command terminator

Command strings sent to the instrument must end with a <Newline> (<NL>) character. IEEE-488 EOI (End or Identify) information can be used as <NL> character to replace termination command string of <NL> character. It is acceptable to place one <NL> after a <Enter>. Termination of command string always resets current SCPI command path to root level.

NOTE

As for every SCPI message with one query sent to the instrument, the instrument will use a <NL> or newline sign (EOI) to terminate response of return. For example, if "DISP:TEXT?" is sent, <NL> will be placed after the returned data string to terminate response. If an SCPI message includes several queries separated by semicolon (such as "DISP?;DISP:TEXT?"), <NL> will terminate response returned after response to the last query. In all cases, the program must read <NL> in response before another command is sent to the instrument, otherwise errors will be caused.

1.6 Data Type

SCPI language defines several data types used for program message and response messages.

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- Numerical parameter

  Commands requiring numerical parameter support the notations of all common decimal notations, including optional signs, decimal points, scientific notation, etc. Special values of numerical parameter are also acceptable, such as MIN, MAX and DEF. In addition, suffixes for engineering units can also be sent together with numerical parameter (including M, k, m or u). If the command accepts only some specific values, the instrument will automatically round the input parameter to acceptable values. The following commands require numerical parameter of frequency value:
  [SOURce[1|2]:]FREQuency:CENTer
  {<Frequency>|MINimum|MAXimum}
  - **<NR1>**: represents an integer value, such as 273;
  - **<NR2>**: represents a real number in floating-point format, such as .273;
  - **<NR3>**: represents a real number in scientific notation, such as 2.73E+2;
  - **<Nrf>**: The extensible form includes <NR1>, <NR2> and <NR3>;
  - **<Nrf+>**: The extensible decimal form includes <Nrf>, MIN, MAX and DEF. MIN and MAX are the minimum and maximum finite number. Within the range of the parameter definition, DEF is the default of the parameter.

- Discrete parameter

  Discrete parameter are used for settings with limited number of programming values (such as IMMediate, EXTernal or BUS). They can use short and long format like key words of commands. They may be expressed in both upper and lower case. The query response always returns uppercase Parameter in short format. The following commands require discrete parameter in voltage unit:
  [SOURce[1|2]:]VOLTage:UNIT {VPP|VRMS|DBM}

- Boolean parameter

  Boolean parameter refer to true or false binary conditions. In case of false conditions, the instrument will accept "OFF" or "0". In case of true conditions, the instrument will accept "ON" or "1". In query of Boolean settings, the instrument will always return "0" or "1". Boolean parameter are required by the following commands:
  DISPlay {OFF|0|ON|1}

- ASCII string parameter

  String parameter may actually include all ASCII character sets. Character strings must start and end with paired quotation marks; and single quotation marks or double quotation marks are both allowed. Quotation mark separators may also act as one part of a string, they can be typed twice without any character added between them. String parameter is used in the following command:
  DISPlay:TEXT <quoted string>
  For example, the following commands display message of "WAITING..." (without quotation marks) on the front panel of the instrument.
  DISP:TEXT "WAITING..."
  Single quotation marks may also be used to display the same message.
  DISP:TEXT 'WAITING...'
  - **<SPD>**: string program data. String parameters enclosed in single or double quotes.
  - **<CPD>**: character program data.

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1.7 Remote Interface Connections

Please refer to user manual for detailed introductions of the remote interface.

Note

If the user want to change the settings of the instrument, for instance, the output setting value, the command **SYST:REM** must be sent to the instrument after finishing the connection between the instrument and PC.

1.8 Multi-channel @chanlist Introduction

Most of the commands in this programming guide support the use of the @Chanlist parameter. When using this parameter, please pay attention to its proper usage to ensure the command takes effect as expected.

For example, in the OUTPut[:STATe] <Bool>[,(@chanlist)] command:

- OUTP 1,(@1,2) turns on the output of both channel 1 and channel 2;
- OUTP 1,(@1) turns on the output of channel 1 only;
- OUTP 1,(@2) turns on the output of channel 2 only;
- OUTP 1 turns on the output of channel 1 by default.

In series or parallel mode, the @chanlist parameter is not required, and the command format should be OUTP 1.

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Chapter 2 BATTery Emulator Subsystem

BATTery:EMULator:VOC:FULL <level>[,(@chanlist)]

Set the full battery voltage of a single cell.

Note: The instructions must be executed when the function is stopped.

Command Syntax

BATTery:EMULator:VOC:FULL <level>[,(@chanlist)]

Arguments

<NRf+>

<MIN~MAX>|MINimum|MAXimum|DEFault

Default Value

5V

Example

Note: Without a channel number, it defaults to operating on channel 1.

BATTery:EMULator:VOC:FULL MAX

BATTery:EMULator:VOC:FULL 10,(@1)

BATTery:EMULator:VOC:FULL MAX,(@2)

BATTery:EMULator:VOC:FULL MAX,(@1:2)

Query Syntax

BATTery:EMULator:VOC:FULL? [(@chanlist)]

Returns

<NRf>

BATTery:EMULator:VOC:EMPTy <level>[,(@chanlist)]

Set the empty battery voltage of a single cell.

Note: The instructions must be executed when the function is stopped.

Command Syntax

BATTery:EMULator:VOC:EMPTy <level>[,(@chanlist)]

Arguments

<NRf+>

<MIN~MAX>|MINimum|MAXimum|DEFault

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Default Value

1V

Example

BATTery:EMULator:VOC:EMPTy 2,(@2)

Query Syntax

BATTery:EMULator:VOC:EMPTy? [(@chanlist)]

Returns

<NRf>

BATTery:EMULator:CAPacity:LIMit <level>[,(@chanlist)]

Set the capacity of a single cell.

Note: The instructions must be executed when the function is stopped.

Command Syntax

BATTery:EMULator:CAPacity:LIMit <level>[,(@chanlist)]

Arguments

<NRf+>

<MIN~MAX>|MINimum|MAXimum|DEFault

Default Value

1Ah

Example

BATTery:EMULator:CAPacity:LIMit MAX,(@2)

Query Syntax

BATTery:EMULator:CAPacity:LIMit? [(@chanlist)]

Returns

<NRf>

BATTery:EMULator:SOC <level>[,(@chanlist)]

Set the initial soc percentage of the simulated battery.

Note: The instructions must be executed when the function is stopped.

Command Syntax

BATTery:EMULator:SOC <level>[,(@chanlist)]

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Arguments

<NRf+>

<MIN~MAX>|MINimum|MAXimum|DEFault

Default Value

50%

Example

BATTery:EMULator:SOC MAX,(@2)

Query Syntax

BATTery:EMULator:SOC? [(@chanlist)]

Returns

<NRf>

BATTery:EMULator:RESistance <level>[,(@chanlist)]

Set the internal resistance of a single cell.

Note: The instructions must be executed when the function is stopped.

Command Syntax

BATTery:EMULator:RESistance <level>[,(@chanlist)]

Arguments

<NRf+>

<MIN~MAX>|MINimum|MAXimum|DEFault

Default Value

0

Example

BATTery:EMULator:RESistance MAX,(@2)

Query Syntax

BATTery:EMULator:RESistance? [(@chanlist)]

Returns

<NRf>

BATTery:EMULator[:STATe] <state>[,(@chanlist)]

Turn on/off the battery simulation function. 0-Stop, 1-Run.

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Command Syntax

BATTery:EMULator[:STATe] <state>[,(@chanlist)]

Arguments

Bool

<0|1|OFF|ON>

Default Value

0|OFF

Example

BATTery:EMULator 1,(@2)

Query Syntax

BATTery:EMULator[:STATe]? [(@chanlist)]

Returns

0|1|2

0-Stop, 1-Run, 2-Pending Trigger

BATTery:EMULator:CURRent:LIMit:POSitive <level>[,(@chanlist)]

Set the positive current limit value of the simulated battery.

Command Syntax

BATTery:EMULator:CURRent:LIMit:POSitive <level>[,(@chanlist)]

Arguments

<NRf+>

<MIN~MAX>|MINimum|MAXimum|DEFault

Default Value

MAX

Example

BATTery:EMULator:CURRent:LIMit:POSitive MAX,(@2)

Query Syntax

BATTery:EMULator:CURRent:LIMit:POSitive? [(@chanlist)]

Returns

<NRf>

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BATTery:EMULator:CURRent:LIMit:NEGative <level>[,(@chanlist)]

Set the negative current limit value of the simulated battery.

Command Syntax

BATTery:EMULator:CURRent:LIMit:NEGative <level>[,(@chanlist)]

Arguments

<NRf+>

<MIN~MAX>|MINimum|MAXimum|DEFault

Default Value

MIN

Example

BATTery:EMULator:CURRent:LIMit:NEGative MIN,(@2)

Query Syntax

BATTery:EMULator:CURRent:LIMit:NEGative? [(@chanlist)]

Returns

<NRf>

BATTery:EMULator:PARallel <num>[,(@chanlist)]

Set the number of single cells connected in parallel.

Note: The instructions must be executed when the function is stopped.

Command Syntax

BATTery:EMULator:PARallel <num>[,(@chanlist)]

Arguments

<NR1>

<MIN~MAX>|MINimum|MAXimum|DEFault

Default Value

1

Example

BATTery:EMULator:PARallel 2,(@2)

Query Syntax

BATTery:EMULator:PARallel? [(@chanlist)]

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Returns

<NR1>

BATTery:EMULator:SERies <num>[,(@chanlist)]

Set the number of single cells connected in series.

Note: The instructions must be executed when the function is stopped.

Command Syntax

BATTery:EMULator:SERies <num>[,(@chanlist)]

Arguments

<NR1>

<MIN~MAX>|MINimum|MAXimum|DEFault

Default Value

1

Example

BATTery:EMULator:SERies 2,(@2)

Query Syntax

BATTery:EMULator:SERies? [(@chanlist)]

Returns

<NR1>

BATTery:EMULator:SAVe <filename>[,(@chanlist)]

Save the currently edited battery simulation test as a file with the specified filename.

Note: The instructions must be executed when the function is stopped.

Command Syntax

BATTery:EMULator:SAVe <filename>[,(@chanlist)]

Arguments

SPD

Default Value

None

Example

BATTery:EMULator:SAVe “battery.csv”,(@2)

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Returns

SRD

BATTery:EMULator:RECall <filename>[,(@chanlist)]

Recall the battery simulation file with the specified filename.

Note: The instructions must be executed when the function is stopped.

Command Syntax

BATTery:EMULator:RECall <filename>[,(@chanlist)]

Arguments

SPD

Default Value

None

Example

BATTery:EMULator:RECall “battery.csv”,(@2)

Returns

SRD

BATTery:EMULator:CAP:INIT <level>[,(@chanlist)]

Set the initial capacity of the battery simulation operation.

Note: The instructions must be executed when the function is stopped.

Command Syntax

BATTery:EMULator:CAP:INIT <level>[,(@chanlist)]

Arguments

<NRf+>

<MIN~MAX>|MINimum|MAXimum|DEFault

Default Value

0.5Ah

Example

BATTery:EMULator:CAP:INIT 5,(@2)

Query Syntax

BATTery:EMULator:CAP:INIT? [(@chanlist)]

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Returns

<NRf>

BATTery:EMULator:VOC:INIT <level>[,(@chanlist)]

Set the initial VOC voltage of the battery simulation operation.

Note: The instructions must be executed when the function is stopped.

Command Syntax

BATTery:EMULator:VOC:INIT <level>[,(@chanlist)]

Arguments

<NRf+>

<MIN~MAX>|MINimum|MAXimum|DEFault

Default Value

3V

Example

BATTery:EMULator:VOC:INIT 6,(@2)

Query Syntax

BATTery:EMULator:VOC:INIT? [(@chanlist)]

Returns

<NRf>

BATTery:EMULator:FILE:COUNt? [(@chanlist)]

Get the number of files.

Command Syntax

BATTery:EMULator:FILE:COUNt? [(@chanlist)]

Arguments

None

Example

BATTery:EMULator:FILE:COUNt? (@2)

Returns

<NR1>

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BATTery:EMULator:FILE:NAME? <index>[,(@chanlist)]

Get the filename of the specified index.

Command Syntax

BATTery:EMULator:FILE:NAME? <index>[,(@chanlist)]

Arguments

<NR1>

1~Total number of files.

Default Value

None

Example

BATTery:EMULator:FILE:NAME? 1,(@2)

Returns

SRD

BATTery:EMULator:CONFig [(@chanlist)]

Apply the simulated battery parameters.

Note: The instructions must be executed when the function is stopped.

Command Syntax

BATTery:EMULator:CONFig [(@chanlist)]

Arguments

None

Example

BATTery:EMULator:CONFig (@2)

Returns

None

BATTery:EMULator:UPDAte [(@chanlist)]

Update the three initial parameters of the simulated battery.

Command Syntax

BATTery:EMULator:UPDAte [(@chanlist)]

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Arguments

None

Example

BATTery:EMULator:UPDAte (@2)

Returns

None

BATTery:EMULator:DELete <filename>[,(@chanlist)]

Delete the simulated battery parameter file with the specified filename.

Note: The instructions must be executed when the function is stopped.

Command Syntax

BATTery:EMULator:DELete <filename>[,(@chanlist)]

Arguments

SPD

Example

BATTery:EMULator:DELete “battery.csv”,(@2)

Returns

None

BATTery:EMULator:MODE <mode>[,(@chanlist)]

Set the simulated battery mode.

Note: The instructions must be executed when the function is stopped.

Command Syntax

BATTery:EMULator:MODE <mode>[,(@chanlist)]

Arguments

CPD

UDEFined|CURVe

Default Value

UDEFined

Example

BATTery:EMULator:MODE CURVe,(@2)

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Query Syntax

BATTery:EMULator:MODE? [(@chanlist)]

Returns

CRD

BATTery:EMULator:SOC:HIGH <level>[,(@chanlist)]

Set the maximum soc percentage of the simulated battery.

Note: The instructions must be executed when the function is stopped.

Command Syntax

BATTery:EMULator:SOC:HIGH <level>[,(@chanlist)]

Arguments

<NRf+>

<0~110>|MINimum|MAXimum|DEFault

Default Value

100

Example

BATTery:EMULator:SOC:HIGH 90,(@2)

Query Syntax

BATTery:EMULator:SOC:HIGH? [(@chanlist)]

Returns

<NRf>

BATTery:EMULator:SOC:LOW <level>[,(@chanlist)]

Set the minimum soc percentage of the simulated battery.

Note: (1)The instructions must be executed when the function is stopped.

(2)The lower limit of SOC must be less than the upper limit. The precondition for this parameter range of -10 to 110 is that the upper limit of SOC is set to 110.

Command Syntax

BATTery:EMULator:SOC:LOW <level>[,(@chanlist)]

Arguments

<NRf+>

<-10~110>|MINimum|MAXimum|DEFault

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Default Value

0

Example

BATTery:EMULator:SOC:LOW 10,(@2)

Query Syntax

BATTery:EMULator:SOC:LOW? [(@chanlist)]

Returns

<NRf>

BATTery:EMULator:MODE:END <mode>[,(@chanlist)]

Set the end mode for the battery simulation.

Note: The instructions must be executed when the function is stopped.

Command Syntax

BATTery:EMULator:MODE:END <mode>[,(@chanlist)]

Arguments

CPD

LAST|LAST_OFF|NORMal

Default Value

NORMal

Example

BATTery:EMULator:MODE:END LAST,(@2)

Query Syntax

BATTery:EMULator:MODE:END? [(@chanlist)]

Returns

CRD

BATTery:EMULator:CURVe <num>,<soc>,<voc>,<res>[,(@chanlist)]

Set the parameters for battery simulation curve.

Note: The instructions must be executed when the function is stopped.

Command Syntax

BATTery:EMULator:CURVe <num>,<soc>,<voc>,<res>[,(@chanlist)]

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Arguments

<NR1>,<NRf+>,<NRf+>,<NRf+>

<MIN~MAX>|MINimum|MAXimum|DEFault,<MIN~MAX>|MINimum|MAXimum|DEFault,<MIN~MAX>|MINimum|MAXimum|DEFault,<MIN~MAX>|MINimum|MAXimum|DEFault

Default Value

1,0,0,0

Example

BATTery:EMULator:CURVe 5,MAX,MAX,MAX,(@2)

BATTery:EMULator:CURVe? <num> [,(@chanlist)]

Query the parameters of battery simulation curve.

Command Syntax

BATTery:EMULator:CURVe? <num> [,(@chanlist)]

Arguments

<NR1>

<MIN~MAX>|MINimum|MAXimum|DEFault

Default Value

1

Example

BATTery:EMULator:CURVe? 5,(@2)

Returns

<NR1>

BATTery:EMULator:RUN:TIME? [(@chanlist)]

Query the running time of the simulated battery.

Command Syntax

BATTery:EMULator:RUN:TIME? [(@chanlist)]

Arguments

None

Example

BATTery:EMULator:RUN:TIME? (@2)

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Returns

<NR1>

BATTery:EMULator:RUN:SOC? [(@chanlist)]

Query the running SOC of the simulated battery.

Command Syntax

BATTery:EMULator:RUN:SOC? [(@chanlist)]

Arguments

None

Example

BATTery:EMULator:RUN:SOC? (@2)

Returns

<NRf>

BATTery:EMULator:RUN:CAPacity? [(@chanlist)]

Query the running capacity of the simulated battery.

Command Syntax

BATTery:EMULator:RUN:CAPacity? [(@chanlist)]

Arguments

None

Example

BATTery:EMULator:RUN:CAPacity? (@2)

Returns

<NRf>

BATTery:EMULator:RUN:CAPacity:POSitive? [(@chanlist)]

Query the upper limit of the running capacity of the simulated battery.

Command Syntax

BATTery:EMULator:RUN:CAPacity:POSitive? [(@chanlist)]

Arguments

None

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Example

BATTery:EMULator:RUN:CAPacity:POSitive? (@2)

Returns

<NRf>

BATTery:EMULator:RUN:CAPacity:NEGative? [(@chanlist)]

Query the lower limit of the running capacity of the simulated battery.

Command Syntax

BATTery:EMULator:RUN:CAPacity:NEGative? [(@chanlist)]

Arguments

None

Example

BATTery:EMULator:RUN:CAPacity:NEGative? (@2)

Returns

<NRf>

BATTery:EMULator:RUN:VOC? [(@chanlist)]

Query the running VOC of the simulated battery.

Command Syntax

BATTery:EMULator:RUN:VOC? [(@chanlist)]

Arguments

None

Example

BATTery:EMULator:RUN:VOC? (@2)

Returns

<NRf>

BATTery:EMULator:RUN:RESistance? [(@chanlist)]

Query the running resistance of the simulated battery.

Command Syntax

BATTery:EMULator:RUN:RESistance? [(@chanlist)]

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Arguments

None

Example

BATTery:EMULator:RUN:RESistance? (@2)

Returns

<NRf>

BATTery:EMULator:TRIG [(@chanlist)]

Trigger the battery simulation function.

Command Syntax

BATTery:EMULator:TRIG [(@chanlist)]

Arguments

None

Example

BATTery:EMULator:TRIG (@2)

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Chapter 3 Example

This chapter displays the programming examples to remotely control IT6600C power supply using SCPI commands.

Note

- If the user want to change the settings of the instrument, for instance, the output setting value, the command **SYST:REM** must be sent to the instrument after finishing the connection between the instrument and PC.
- " - >" indicates the commands that you send to the IT6600C power supply.

Example: BATTery:EMULator

UDEFined

SYST:REM

BATTery:EMULator:MODE UDEFined,(@2)

BATTery:EMULator:CAPacity:LIMit MAX,(@2)

BATTery:EMULator:VOC:FULL MAX,(@2)

BATTery:EMULator:VOC:EMPTy 2,(@2)

BATTery:EMULator:PARallel 1,(@2)

BATTery:EMULator:SERies 1,(@2)

BATTery:EMULator:RESistance 1,(@2)

BATTery:EMULator:SOC:HIGH 95,(@2)

BATTery:EMULator:SOC:LOW 5,(@2)

BATTery:EMULator:CURRent:LIMit:POSitive MAX,(@2)

BATTery:EMULator:CURRent:LIMit:NEGative MIN,(@2)

BATTery:EMULator:SOC 50,(@2)

BATTery:EMULator:VOC:INIT 3,(@2)

BATTery:EMULator:CAP:INIT 1,(@2)

BATTery:EMULator:MODE:END LAST,(@2)

BATTery:EMULator:SAVe "battery.csv",(@2)

BATTery:EMULator:CONFig (@2)

BATTery:EMULator:UPDAte (@2)

OUTPut 1

BATTery:EMULator 1,(@2)

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BATTery:EMULator:TRIG (@2)

CURVe

SYST:REM

BATTery:EMULator:MODE CURVe,(@2)

BATTery:EMULator:CAPacity:LIMit MAX,(@2)

BATTery:EMULator:SOC:HIGH 95,(@2)

BATTery:EMULator:SOC:LOW 5,(@2)

BATTery:EMULator:CURRent:LIMit:POSitive MAX,(@2)

BATTery:EMULator:CURRent:LIMit:NEGative MIN,(@2)

BATTery:EMULator:SOC 50,(@2)

BATTery:EMULator:CAP:INIT 1,(@2)

BATTery:EMULator:MODE:END LAST,(@2)

BATTery:EMULator:CURVe 1,90,20,0.8,(@2)

BATTery:EMULator:CURVe 2,80,18,0.9,(@2)

BATTery:EMULator:CURVe 3,70,16,0.9,(@2)

BATTery:EMULator:SAVe “battery.csv”,(@2)

BATTery:EMULator:CONFig (@2)

BATTery:EMULator:UPDAte (@2)

OUTPut 1

BATTery:EMULator 1,(@2)

BATTery:EMULator:TRIG (@2)

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